One-Hot-Preserving QAOA for Flood Shelter Assignment: Pairwise Capacity Encoding and Angle Transfer

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***Abstract*—Flood shelter assignment is a small capacity-constrained assignment problem that is often proposed for quantum optimization. We study the Quantum Approximate Optimization Algorithm (QAOA) on instances from a flood digital twin by exact classical simulation. Capacity enters as approximate pairwise penalties instead of slack qubits, so a classical verifier checks every sampled plan. At 12 qubits and depth 4, a ring-XY mixer that keeps each group assigned to one shelter placed a mean probability of 0.188 on the optimal plan, against at most 0.048 for the transverse-field mixer at the same CNOT count. We then fixed one angle set per depth, chosen on development instances by mean optimal-plan probability, a rule that needs their optima. Applied unchanged, it reached 0.306 on held-out seeds against 0.163 for per-instance training, and it stayed ahead on 150 larger instances of up to 27 qubits. Rules that need no optima were not reliable in leave-one-out tests, and transfer failed without a shared energy scale. The optimal-plan probability still fell about 2.9-fold per added group, as capacity pushed most of the probability onto infeasible plans. On road networks of 6 Indian cities, an exact penalty over minimal overloading sets raised the feasible probability from 0.601 to 0.922 but not the optimal-plan probability, at 66 instead of 28 CNOTs per layer. Exhaustive search solves the 12-qubit instances in about a millisecond; we claim no quantum advantage.**

***Index Terms*—QAOA, XY mixer, parameter transfer, QUBO, evacuation planning**

I. Introduction

When a flood closes roads and fills shelters, an emergency officer must decide again which shelter each evacuation group should go to, trading off travel time, flood exposure and the seats left. As a binary program this is a capacity-constrained assignment problem, the kind that is routinely encoded as a quadratic unconstrained binary optimization (QUBO) problem [1], [2] and offered as an application for near-term quantum optimization [3], [4].

Prior proposals rarely ask three questions. Does the circuit respect the constraints or only penalise violations? The original QAOA [5] explores all 2^n bitstrings, most of which are not plans, while the quantum alternating operator ansatz [6] keeps the state in a feasible subspace [7]–[9]; but mixer studies treat equality constraints, and capacity is an inequality. Must the angles be retrained after every disruption, and how should a reusable set be chosen? And how do the results compare with classical methods and with a sampler followed by a feasibility check?

We address them on instances generated by the digital twin of a flood decision-support prototype. Our contributions are:

1) A shelter-assignment QUBO in which capacity is encoded by pairwise products instead of slack qubits, with a cost shift and scaling that place all instances on one energy range; we show that it admits infeasible zero-penalty plans and compare it with an exact penalty over minimal overloading sets, integer programming, exhaustive search, a greedy rule and simulated annealing.
2) A comparison of transverse-field, ring-XY, Grover and feasible-subspace mixers under mean-energy and conditional value-at-risk (CVaR) training over 5 optimizer seeds, with Holm-corrected tests and per-layer CNOT counts.
3) An angle-transfer study with three selection rules, two of which need no optima, an ablation in which transfer failed without the shared energy scale, evidence that trained sets concentrate across instances, and a size test of 12-qubit angles up to 27 qubits.

II. Related Work

Formulations. Lucas gave Ising forms for many NP-hard problems [1], and Glover *et al.* describe how assignment and

capacity constraints become quadratic penalties, with slack variables for inequalities [2]. Flood shelter allocation has been solved by integer programming [10] and by quantum annealing [3].

Mixers. QAOA alternates a cost unitary and a mixing unitary for p layers [5], [11]. Hadfield *et al.* generalised the mixer to preserve hard constraints [6]. Wang *et al.* analysed XY mixers for one-hot constraints [7], Fuchs *et al.* built constraint-preserving [8] and subspace mixers [12], and Grover mixers move the complexity into state preparation [9]. XY mixers have also served routing [13] and rostering [14].

Training and transfer. Angles can be interpolated across depths [15], CVaR objectives favour good samples [16], and warm starts use classical solutions [17]. The objective value at fixed angles concentrates across typical instances [18], optimal angles transfer between random graphs [19], and for weighted MaxCut a rescaling of the weights lets one parameter set serve many instances [20]. On hardware, QAOA performance falls with size and depth [21], and practical advantage has not been shown [4].

Positioning. The ring-XY mixer is that of [7], and the transfer studies [18]–[20] use unconstrained MaxCut with the standard mixer. We add a capacity inequality, as a pairwise or exact penalty, next to a one-hot-preserving mixer, compare selection rules with and without optima, and test on held-out instances and real road networks with a verifier in the loop.

III. Problem Formulation

A. Instances from the Digital Twin

The twin models a corridor with three population zones, nine junctions and three shelters joined by 22 roads. Every road carries a travel time, a flood-depth index and a congestion level drawn from a seed. Shelters have capacities of 120, 90 and 70 people with an initial occupancy of 15–45%, and groups have 8–22 people. Each instance applies the disruption used in the twin's routing benchmark: road R5 closes and Shelter A receives arrivals equal to half its capacity. For group g and shelter s we take the best open path under the twin's flood- and congestion-weighted edge cost and define

$$c_{gs} = T_{gs} + 5H_{gs}, \tag{1}$$

where T_{gs} is travel time in minutes and H_{gs} the mean flood index along the path. The pair (g, s) is *forbidden* if no open path exists, the shelter has failed, or the group alone exceeds the remaining seats R_s. With G groups, binary variables x_{gs} give $n = 3G$ qubits. We write m_g for the number of people in group g. A plan is feasible if every group has exactly one shelter, no forbidden pair is used and

$$\textstyle\sum_g m_g x_{gs} \le R_s \quad \text{for each shelter } s. \tag{2}$$

B. QUBO with Pairwise Capacity

An inequality normally needs slack qubits [2]. We use a pairwise encoding instead. Let $\mathcal{O} = \{(g, h, s) : m_g + m_h > R_s\}$ be the pairs of groups that cannot share shelter s and $\mathcal{F}$ the forbidden pairs. With shifted costs $c'_{gs} = c_{gs} - \min_{s'} c_{gs'}$ and penalty weight μ,

$$E(\mathbf{x}) = \sum_{g,s} c'_{gs} x_{gs} + \mu\Big[\sum_g \Big(1 - \sum_s x_{gs}\Big)^2 + \sum_{(g,s)\in\mathcal{F}} x_{gs} + \sum_{(g,h,s)\in\mathcal{O}} x_{gs}x_{hs}\Big]. \tag{3}$$

Remark 1: With $\mu = 1 + \sum_g \max_s c'_{gs}$, every bitstring that breaks a one-hot, forbidden-pair or pairwise-capacity term has higher energy than every bitstring that breaks none of them, since $c'_{gs} \ge 0$ bounds the cost of a bitstring without violations by $\mu - 1$ and each violation adds at least μ. The bound no longer holds for the reduced weights $\lambda\mu$ with $\lambda < 1$ used in the penalty sweep of Section VI-A.

On one-hot states the shift subtracts the constant $\sum_g \min_s c_{gs}$, so the optimal plan is unchanged. We divide the Hamiltonian by $\Delta = \mu - 1$, which maps the energies of all penalty-free plans into $[0, 1]$ for every instance. The pairwise encoding misses overloads by three or more groups whose pairs all fit, so a penalty-free plan can still be infeasible, and every sampled plan is re-checked exactly before use.

An exact penalty also needs no slack qubits. Let $\mathcal{M}$ hold the minimal overloading sets (T, s): groups T allowed at s with $\sum_{g\in T} m_g > R_s$, while every proper subset fits. Replacing the last sum in (3) by

$$\textstyle\sum_{(T,s)\in\mathcal{M}} \prod_{g\in T} x_{gs} \tag{4}$$

leaves no capacity penalty exactly when no shelter is overloaded, because an overloaded shelter always holds a minimal set; the pairs of $\mathcal{O}$ are the sets of size two. A product of k variables expands into $2^k - 1$ Pauli-Z strings, and a string of weight w costs $2(w - 1)$ CNOTs.

IV. QAOA Variants and Training

For depth p and angles $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta})$, QAOA prepares

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k \hat{E}/\Delta}\, |\psi_0\rangle, \tag{5}$$

where $\hat{E}$ is diagonal with entries $E(\mathbf{x})$. Substituting $x_q = (1 - Z_q)/2$ turns (3) into an Ising Hamiltonian with one R_Z rotation per qubit and one ZZ rotation, two CNOTs, per non-zero coupling.

Transverse-field (X) mixer. $B = \sum_q X_q$ and $|\psi_0\rangle = |+\rangle^{\otimes n}$; constraints act only through the penalties. A cost layer has 12 one-hot couplings and on average 2.0 capacity couplings, 28 CNOTs; the mixer needs none.

Ring-XY mixer. For each group g with qubits a_g, b_g, c_g,

$$B_{\mathrm{XY}} = \sum_g \sum_{(u,v)\in\{(a_g,b_g),(b_g,c_g),(c_g,a_g)\}} \tfrac{1}{2}\big(X_u X_v + Y_u Y_v\big). \tag{6}$$

Each term swaps amplitude between $|01\rangle$ and $|10\rangle$, so a one-hot group stays one-hot [7]. The start state $|W_3\rangle^{\otimes G}$, with $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$, is the uniform superposition of the 3^G one-hot plans [22]. The one-hot penalty is

then constant and dropped; the three ring terms are applied in a fixed order. Each XY rotation compiles to two CNOTs [23], so a layer has 28 CNOTs, the same count as the X variant. We prepare $|W_3\rangle$ with an R_Y rotation, a controlled Hadamard (one CNOT) and two CNOTs, which we checked numerically; this adds 12 CNOTs once.

Grover mixer. $B_{\mathrm{GM}} = |\psi_0\rangle\langle\psi_0|$ with the same start state, so $e^{-i\beta B_{\mathrm{GM}}} = I + (e^{-i\beta} - 1)|\psi_0\rangle\langle\psi_0|$ [9]. A layer needs the inverse preparation, a phase on $|0\rangle^{\otimes n}$ controlled by all n qubits, and the preparation. With the textbook ancilla construction [24] the controlled phase takes $2(n-2)$ six-CNOT Toffolis, one controlled phase and $n-2=10$ ancillas (122 CNOTs), so a Grover layer has 150 CNOTs.

Feasible-subspace mixer. As a reference bound, not a usable circuit, a Grover mixer starts from the uniform superposition of the feasible plans, which requires the feasible set [9], [12].

Objectives and training. We minimise either $\langle\hat{E}\rangle/\Delta$ or the CVaR with $\alpha = 0.1$ [16], the only level we tried; it was not tuned. Sorting bitstrings by energy, $E_{(1)} \le E_{(2)} \le \dots$, with probabilities $\pi_{(k)}$ and cumulative mass $\Pi_k = \sum_{j\le k}\pi_{(j)}$,

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha}\sum_k E_{(k)}\,\min\big(\pi_{(k)},\ \max(0,\ \alpha - \Pi_{k-1})\big). \quad (7)$$

Every variant is trained from scratch with the same procedure: COBYLA [25] with at most 250 evaluations per start, depth by depth. At each depth p we start from two random points ($\gamma_k \sim U(0,\pi)$, $\beta_k \sim U(0,\pi/2)$) and, for $p>1$, from the optimum of depth $p-1$ interpolated to p layers [15], and keep the start with the lowest objective; training never uses the optimum.

A. Angle Transfer

Since the scaled Hamiltonians of all instances share the energy range $[0,1]$, we test whether trained angles carry over. Let $\Theta_p = \{\boldsymbol{\theta}_j^{(p)} : j \in \mathcal{D}\}$ be the angles trained on the development instances $\mathcal{D}$. We fix one set $\boldsymbol{\theta}_p^\star \in \Theta_p$ by one of three rules:

$$\boldsymbol{\theta}_p^\star = \arg\max_{\boldsymbol{\theta}\in\Theta_p}\ \tfrac{1}{|\mathcal{D}|}\textstyle\sum_{j\in\mathcal{D}} P_{\mathrm{opt}}(j;\boldsymbol{\theta}), \quad (8)$$

or the set with the lowest mean of $\langle\hat{E}\rangle/\Delta$, or with the lowest mean CVaR, over $\mathcal{D}$. Rule (8) needs the exact optimum of every development instance, from exhaustive search; it is an oracle on the development set, though it uses nothing from the instances it is applied to. The other two rules use only quantities a device can estimate. We choose among whole trained sets because their coordinate-wise median, which mixes basins of the periodic angle space, gave $P_{\mathrm{opt}} = 0.001$ in leave-one-out.

V. Experimental Setup

Instances. The development set has 30 instances from twin seeds $42+17i$, $i=0,\dots,29$, with the first four groups ($G=4$, $n=12$). All are feasible, with on average 45.0 feasible plans among the 81 one-hot plans, and in 12 of them capacity changes the optimal plan (*capacity-binding*). The held-out set has 30 further seeds, $i=30,\dots,59$, of the same disruption pattern, used only for the final transfer test. We also built 60 instances (50 feasible, 22 capacity-binding) on the OpenStreetMap road graphs of 6 flood-prone Indian cities, with each city's mapped zones, shelters and road flood-exposure proxy. Group sizes and seats follow the twin, as demand is not mapped; the disruption closes the most flood-exposed road on a zone–shelter path and fills half of the first shelter's capacity.

Metrics. Simulations store all 2^n amplitudes in double precision, with fixed seeds; code and stored results will be released after review. P_{opt} is the probability of measuring an optimal plan, a feasible plan of lowest true cost (not always the minimiser of (3)), and P_{feas} that of a feasible plan. A sampler draws N plans and returns the cheapest one that passes an exact feasibility check. It returns the optimum whenever the optimum is among the samples, which has probability success@$N = 1-(1-P_{\mathrm{opt}})^N$, assuming independent shots. We also report the expected relative gap $(c-c^\star)/c^\star$ of the returned plan, given that a feasible plan was drawn, for $N \in \{8, 32, 128\}$, computed exactly from the output distribution. An evaluation is one estimate of the training objective (on hardware, a batch of circuit runs).

Statistics. Paired comparisons use the two-sided Wilcoxon signed-rank test on per-instance values, also for success@N. We apply the Holm correction within each family of tests: the four depths of one mixer comparison, the 7 rows of Table II, the four city comparisons, the 10 size comparisons and the four tests of Table IV. Effect sizes are median paired differences. Intervals for variants run with 5 optimizer seeds use a two-level bootstrap that resamples instances and then one seed per instance. Because the city instances cluster by city, we also apply a sign test to the per-city mean differences.

Baselines. Uniform sampling over the 81 one-hot plans ($P_{\mathrm{opt}} = 0.0169$) and over the feasible plans ($P_{\mathrm{opt}} = 0.049$); exhaustive search; the integer program solved by HiGHS [26]; a greedy rule that places the largest group first in its cheapest shelter with seats left; and simulated annealing (SA) [27] with 20 restarts of 200 sweeps, either with bit flips on (3) or with moves that reassign one group, which keep the plan one-hot. For the X mixer we swept the penalty $\lambda\mu$ with $\lambda \in \{0.25, 0.5, 1, 2\}$.

VI. Results

A. Mixers, Objectives and Penalties

Table I and Fig. 1 compare the mixers. Over 5 seeds the XY mixer reached $P_{\mathrm{opt}} = 0.188 \pm 0.020$ at $p=4$ (seed means 0.168–0.215), against 0.035–0.048 for the X mixer. It was better on 25–29 of 30 instances at every depth (Holm-adjusted $p \le 8.0\times10^{-6}$), and the ranking held for every seed. Its feasibility stayed between 0.722 and 0.795 at every depth, whereas the X mixer began with $P_{\mathrm{feas}} = 0.074$ and spent its layers recovering feasibility. As every sample is verified, the fairer figure is the chance that a feasible sample is optimal, $P_{\mathrm{opt}}/P_{\mathrm{feas}}$: at $p=4$ it was 0.28 for XY against 0.08 for X.

TABLE I
MIXERS AND OBJECTIVES AT DEPTH 4 (30 DEVELOPMENT INSTANCES, 12 QUBITS)

Variant	P_{opt}	95% CI	P_{feas}	s@32 %	Evals	CNOT
X, mean	0.040	0.028–0.055	0.569	60.8	2,091	28
XY, mean	0.188	0.127–0.257	0.755	82.5	2,510	28
XY, CVaR	0.139	0.117–0.166	0.664	97.7	1,356	28
Grover, mean	0.139	0.121–0.157	0.815	96.9	2,033	150
Grover, CVaR	0.117	0.106–0.127	0.563	96.2	918	150
Feasible subspace	0.444	0.385–0.511	1.000	100.0	2,039	n/a

X, XY: mean over 5 seeds, CI by two-level bootstrap. Evals: objective evaluations per instance, depths 1–4. CNOT: per layer, all-to-all connectivity.

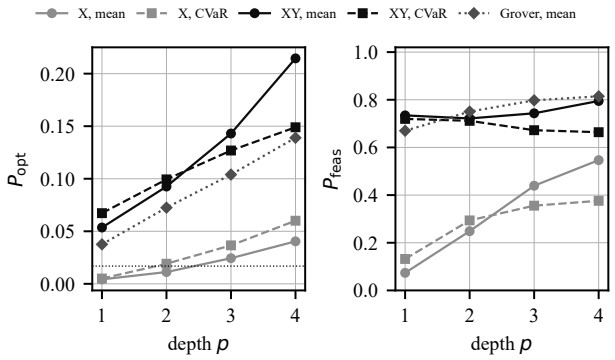


Fig. 1. Probability of the optimal plan (left, different y-range) and of any feasible plan (right) against depth, optimizer seed 0. Dotted: uniform one-hot.

The Grover mixer reached 0.139 at $p = 4$, and its difference from XY was not significant at any depth (Holm-adjusted $p \geq 0.19$). The data lean towards XY: at $p = 4$ it was ahead on 20 of 30 instances, with a mean difference of 0.076 (paired bootstrap interval 0.018–0.141) and an unadjusted $p = 0.047$. Its success@32 was higher, but a layer costs 150 CNOTs and 10 ancillas against 28 CNOTs.

CVaR training lowered the XY mixer's mean P_{opt} but raised success@32 from 88.4% to 98.1% (seed 0): the worst instance (capacity-binding, like all the worst) rose from 0.002 to 0.074.

A smaller penalty helped the X mixer only partly: lowering the weight to $\lambda = 0.25$ raised its P_{opt} by 2.6× to 0.104 at a feasibility of only 0.401, still about half of the XY result. As a bound on the cost of a capacity penalty, a mixer restricted to feasible plans reached $P_{\text{opt}} = 0.444$ at $p = 4$, 9.0 times uniform sampling over the same feasible plans, and beat XY on 26 of 30 instances ($p = 4.4 \times 10^{-6}$).

B. Angle Transfer to Held-Out Seeds

Table II gives the held-out test. For the XY mixer with the mean objective, the set chosen by (8) reached $P_{\text{opt}} = 0.306$ against 0.163 for per-instance training with 2,496 evaluations (better on 24 of 30, median difference 0.195, Holm $p = 8.5 \times 10^{-4}$). The oracle-free sets also beat per-instance training on these seeds: 0.264 for the energy rule ($p = 3.8 \times 10^{-3}$) and 0.234 for the CVaR rule ($p = 0.039$). Leave-one-out on the development set did not confirm this: the energy rule gave 0.082 and the CVaR rule 0.164, both below the 0.215 of per-instance training, while rule (8) gave 0.292. The same held for CVaR angles chosen by CVaR (0.250 on held-out seeds, 0.095 in leave-one-out against 0.149). An oracle-free choice therefore gave no reliable gain, while rule (8) gained in both tests. Transfer did not rescue the X mixer (0.036).

Part of the gap is a mismatch of objectives: per-instance training reached a lower mean $\langle \hat{E} \rangle / \Delta$ than the fixed set (0.300 against 0.530, lower on 24 of 30), but across the 30 trained sets mean energy and mean P_{opt} had a rank correlation of only 0.33. Stronger optimizers stayed below: best of five COBYLA seeds 0.203 (12,485 evaluations), L-BFGS-B 0.225. COBYLA started from the fixed angles did improve on them (0.385 with 814 evaluations, $p = 1.9 \times 10^{-3}$).

A fixed set can work because performance concentrates across instances: a trained set's mean P_{opt} on the development set predicted its mean on the held-out seeds with rank correlation 0.98. Only 7 of the 30 trained sets beat per-instance training there (median 0.062), so the choice matters. The gain came mostly from capacity-binding instances: on the 16 binding seeds per-instance training reached 0.106 and the fixed set 0.312, on the others 0.228 and 0.298. We repeated training, selection and the held-out test without the scaling and with division by the standard deviation of the penalty-free energies. Without scaling the fixed angles reached only 0.029, although per-instance training still reached 0.270; with the other normalisation they reached 0.282, close to our 0.306. Transfer therefore needs a common energy scale, and the two scales we tried served about equally well. The one-time cost, 75,392 training and 3,600 selection evaluations, equals per-instance training on about 32 disruptions.

C. Sampling with a Feasibility Check

On the held-out seeds at $p = 4$, the best verified plan among 8 samples came within 0.09% of the optimal cost on average for the fixed XY angles, 1.49% for per-instance training and 6.7% for uniform one-hot sampling; with 128 samples per-instance training was within 0.10%. As many plans cost close to the optimum, success@N is the stricter measure.

D. Transfer to Real Road Networks

With angles fixed on the twin, pooled over the 50 instances, the set chosen by (8) reached $P_{\text{opt}} = 0.202$ against 0.213 for per-instance training (Holm $p = 0.74$) and the energy-rule set 0.216 ($p = 0.38$). It led in only 3 of 6 cities (sign test $p = 1.00$), so there was no gain in P_{opt}. Per-instance training left 8 instances with $P_{\text{opt}} < 0.01$, the fixed set 1, and success@32 rose from 74.9% to 89.9% (Holm $p = 0.042$) and to 96.7% for the energy rule ($p = 9.7 \times 10^{-5}$), in all 6 cities (sign test $p = 0.03$).

E. Larger Instances and Gate Noise

The ring-XY circuit never leaves the 3^G one-hot amplitudes, so we simulated it exactly in that subspace; on 40 random 12- and 15-qubit circuits it agreed with the full statevector to 1.6×10^{-16}. We took 30 new twin instances for each of $G = 5, \ldots, 9$ groups (15 to 27 qubits) with the same

TABLE II
ANGLES AT $p = 4$ ON THE 30 HELD-OUT SEEDS: PER-INSTANCE TRAINING AGAINST ONE FIXED ANGLE SET CHOSEN ON THE DEVELOPMENT SET

Variant	Angles	Needs optima	Held-out P_{opt}	Median paired diff.	Wins of 30	Holm p	Leave-one-out P_{opt} (dev.)	s@32 %
XY, mean	per instance	no	0.163				0.215	79.7
	best of 5, min. energy	no	0.203				0.245	84.9
	L-BFGS-B	no	0.225					76.9
	fixed, max. P_{opt}	yes	0.306	+0.195	24	8.5×10^{-4}	0.292	100.0
	fixed, min. energy	no	0.264	+0.131	24	3.8×10^{-3}	0.082	99.9
	fixed, min. CVaR	no	0.234	+0.102	23	0.039	0.164	99.8
XY, CVaR	per instance	no	0.138				0.149	97.9
	fixed, max. P_{opt}	yes	0.250	+0.070	29	6.5×10^{-8}	0.259	99.8
	fixed, min. CVaR	no	0.250	+0.070	29	6.5×10^{-8}	0.095	99.8
X, mean	per instance	no	0.028				0.040	50.3
	fixed, max. P_{opt}	yes	0.036	+0.010	20	0.039	0.039	63.6
	fixed, min. energy	no	0.034	+0.006	21	0.064	0.034	65.8

Fixed sets vs. per-instance training of the same variant (Wilcoxon signed-rank, Holm over 7 rows). Per-instance rows: development-set mean in the leave-one-out column.

TABLE III
FIXED 12-QUBIT ANGLES ON 30 TWIN INSTANCES PER SIZE n (QUBITS), XY MIXER, $p = 4$; FIX-O: SET OF RULE (8), FIX-E: ENERGY RULE; P_{feas} OF FIX-O

	P_{opt}						s@32 %	
n	trained	fix-O	Holm p	fix-E	Holm p	P_{feas}	trained	fix-O
15	0.1110	0.1924	4.0×10^{-3}	0.1470	0.083	0.50	56.7	99.6
18	0.0415	0.0670	0.027	0.0505	0.083	0.36	31.2	84.4
21	0.0037	0.0211	2.0×10^{-5}	0.0139	1.5×10^{-3}	0.13	9.3	42.5
24	0.0016	0.0047	4.0×10^{-3}	0.0034	5.7×10^{-3}	0.04	3.9	12.4
27	0.0013	0.0026	0.011	0.0012	0.083	0.05	3.5	7.6

TABLE IV
PAIRWISE VS. EXACT CAPACITY PENALTY, 50 CITY INSTANCES, $p = 4$. MISS: INSTANCES WITH AN INFEASIBLE PENALTY-FREE PLAN; MIN.: MINIMISER OPTIMAL

				trained		fix-O	
Penalty	miss	min. %	CNOT	P_{feas}	P_{opt}	P_{feas}	P_{opt}
Pairwise	47	74	28	0.601	0.219	0.571	0.202
Exact	0	100	66	0.922	0.169	0.675	0.177
Holm p				1.5×10^{-6}	0.081	5.4×10^{-3}	0.034

disruption, applied the 12-qubit sets of rule (8) and of the energy rule unchanged, and trained every instance with the procedure above (Table III). The set of rule (8) beat per-instance training at every size (Holm $p \leq 0.027$ over 10 tests). The energy-rule set won on at least 22 of 30 instances at every size but was significant only at 21 and 24 qubits. Per-instance training (about 2,438 evaluations) fell below $P_{\text{opt}} = 0.01$ on at least 28 of 30 instances from 21 qubits on.

In absolute terms, the fixed set's mean P_{opt} fell by a factor of about 2.9 per added group, to 0.003 at 27 qubits, still a median 16 times uniform one-hot sampling; from 21 qubits on it fell below uniform sampling on up to 4 instances. Capacity explains most of the drop. From 24 qubits on every instance was capacity-binding, and the fixed circuit put at most 0.053 of its probability on feasible plans, since the one-hot mixer sees capacity only through the penalty, so most samples fail the verifier.

In a trajectory noise model (200 runs per circuit, 10 development instances), each two-qubit gate fails with probability ϵ and applies a random non-identity two-qubit Pauli. As each ZZ and XY gate compiles to two CNOTs, $\epsilon = 0.01$ means a CNOT error near 0.5%, close to the 0.36–0.62% two-qubit errors of a superconducting processor [28]. With angles trained per instance, at $\epsilon = 0.01$ and $p = 4$ the XY mixer kept $P_{\text{opt}} = 0.105$, 3.2× the X mixer's 0.033. A global depolarising model gave a lower 0.086, as many Pauli errors barely change the output.

F. Encoding Exactness and Classical Baselines

In all 30 development instances some infeasible one-hot plan had zero penalty: three groups whose pairs fit overloaded a shelter together. Such plans usually cost more than the optimum, so the minimiser of (3) was the true optimum in 93.3% of instances. Exhaustive search took 0.90 ms per instance. The greedy rule found a feasible plan in 100.0% of instances and the optimum in 93.3%. Bit-flip SA ended at the optimum in 80.0% of instances and one-hot SA in 93.3%, the encoding's limit; both visited the optimum every time, so keeping the cheapest verified plan found it in 100%.

On the 50 city instances, the minimal-set penalty (4) was exact on every instance, checked over all 81 one-hot plans, while the pairwise penalty admitted 262 infeasible penalty-free plans in 47 instances and its minimiser was optimal in only 74%. Exhaustive search took 0.39 ms per instance and HiGHS 2.6 ms; both returned the optimum every time. Table IV trains ring-XY on both penalties. The exact one raised P_{feas} from 0.601 to 0.922 (Holm $p = 1.5 \times 10^{-6}$), in all 6 cities, but not P_{opt} (0.219 against 0.169, $p = 0.081$), and success@32 went from 75.6% to 81.2% ($p = 0.91$). It helped the 22 capacity-binding instances (P_{opt} 0.061 to 0.125) and hurt the others (0.342 to 0.203). Its higher-order terms cost 66 CNOTs per layer (up to 120) against 28, and a slack encoding would add 15 to 19 qubits to the 12.

VII. DISCUSSION AND LIMITATIONS

Three findings matter for hardware experiments. First, the one-hot constraint belongs in the mixer: at the same CNOT

count, ring-XY beat the transverse-field mixer on almost every instance and seed. Capacity is harder. The feasible-subspace mixer doubled P_{opt} but needs the feasible set; once capacity bound on every instance, ring-XY put almost all its probability on infeasible plans; and the exact penalty fixed feasibility at 12 qubits but not P_{opt}, for more than twice the CNOTs. A capacity-aware mixer is the next step. Second, transfer needs a common energy scale and a good selection rule: with development-set optima one fixed set beat per-instance training for a one-time cost of 78,992 evaluations, but without optima the gain was not reliable. Third, the pairwise penalty admits infeasible penalty-free plans, so every sample must be verified.

The exact penalty was tested on the 12-qubit city instances only. All twin instances share one disruption pattern, and the city instances use sampled demand. The noise model charges errors per logical gate, ignores single-qubit and correlated errors and was run at 12 qubits; no circuit was compiled for hardware. P_{opt} depends on the training budget, and the Grover test cannot prove equivalence.

VIII. Conclusion

Ring-XY gave QAOA a significantly higher P_{opt} than the transverse-field mixer at equal CNOT count, and angles fixed with development-set optima beat per-instance training up to 27 qubits. Capacity is the open problem: the pairwise penalty needs a verifier, and an exact penalty raised feasibility but not P_{opt}. The results guide hardware experiments and show no present-day advantage.

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